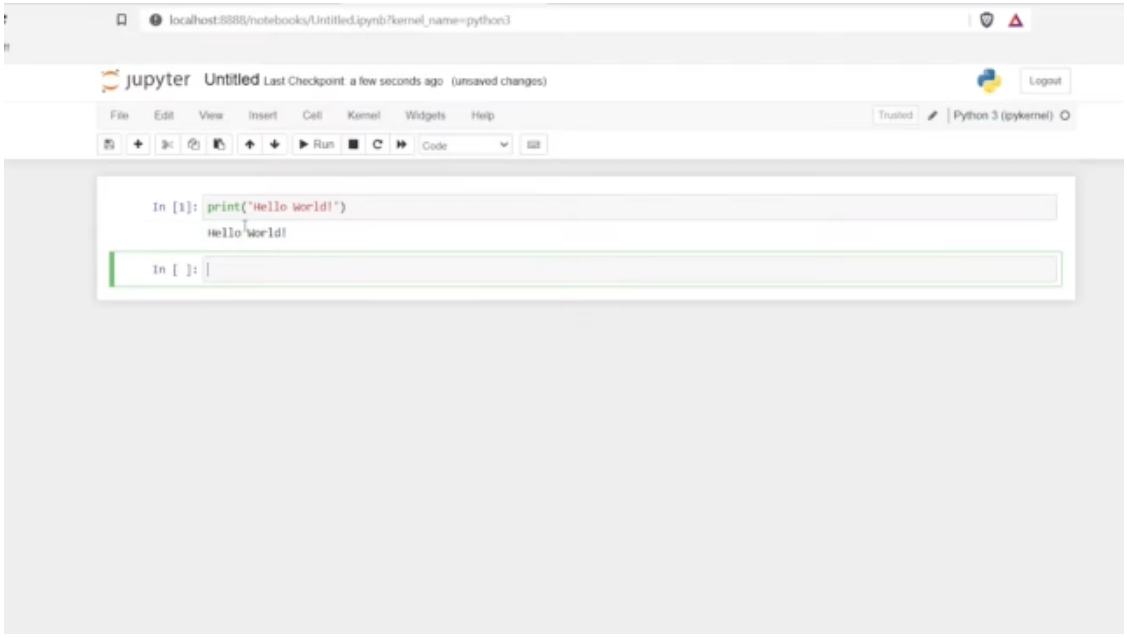
Frame Review: Keep the Lesson

Keep the code. Calm the blink.
Joseph Ayanda / AI-assisted development / OpenCV AI Competition 2026 / 23 October



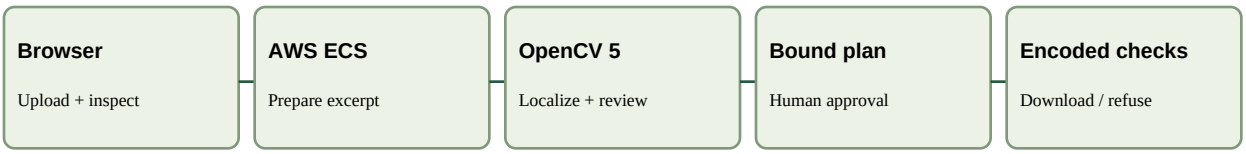
A real Jupyter lesson by Alex The Analyst, via Wikimedia Commons (CC BY 3.0 as listed; imported YouTube license review not yet reviewed). Seconds 301-313 were extracted, resized and re-encoded. The caret already occurs in the original. No added defect or author endorsement.

The user and decision. A lesson editor has a flattened recording and no editable project. The code has been written, yet a caret keeps blinking during the explanation. Frame Review lets the editor hold one actual observed state in a reviewed box and interval, while code, narration and timing remain checked against the exported file. A caret can also be useful teaching information; the editor must compare both states and decide.

Current measured evidence	Scope
Six untouched original sources complete HTTP repair	Three new cohort sources, two development repairs, one previously qualified source repeated.
41 internal changes -> 0; fixed blur retains 41	Six selected four-second intervals; count excludes entry and return.
Teaching pixels, prepared audio and timing pass	Independent native/RGB/PCM checks; all 360 teaching frames per case.
Jupyter and npm originals pass public AWS browser flows	Downloads, twelve crops, reload and downloaded playback; same sources, not extra successes.

No creator time saving, attention, comprehension, accessibility, adoption or safety benefit has been measured. The contribution is a functioning, inspectable repair with narrow preservation evidence.

Vision proposes. The editor decides.



Managed HTTPS / digest-pinned ECR / Fargate CPU / CloudWatch Logs

FFmpeg and OpenCV run on AWS. No model API or application AWS role is required.

One worker / eight slots / owned ephemeral jobs / original-source recovery

Substantive OpenCV 5. Connected components localize small stationary support. Native luma measurements test two observed states, repeated transitions, state separation, bounded appearance error and chroma stability. OpenCV also creates overlays/crops and measures encoded RGB error histograms in named teaching areas. NumPy represents YUV samples and mappings. FFmpeg handles bounded preparation, lossless export and prepared-audio transport.

The reviewed action. steady-detail-v1 holds an observed state with greater contrast against its surrounding ring. It does not generate replacement imagery or identify semantic distractions. A source-bound plan names the exact box, frames, observed reference and teaching rectangles. Approval binds both analyses, implementation and source; the worker recomputes the proposal before export. Empty or overlapping teaching protection, changed inputs, stale plans and failed encoded checks withhold the edited download.

Exact preservation. Checks cover native samples outside the box, the intended mapping inside it, all 240 outside-interval frames, every decoded RGB sample outside the four-pixel influence area, and every declared teaching sample in all 360 frames. Prepared AAC packet bytes, timestamps, skip metadata and decoded PCM must match. The uploaded original stays separately hash-bound and recoverable. No-edit decisions remain downloadable.

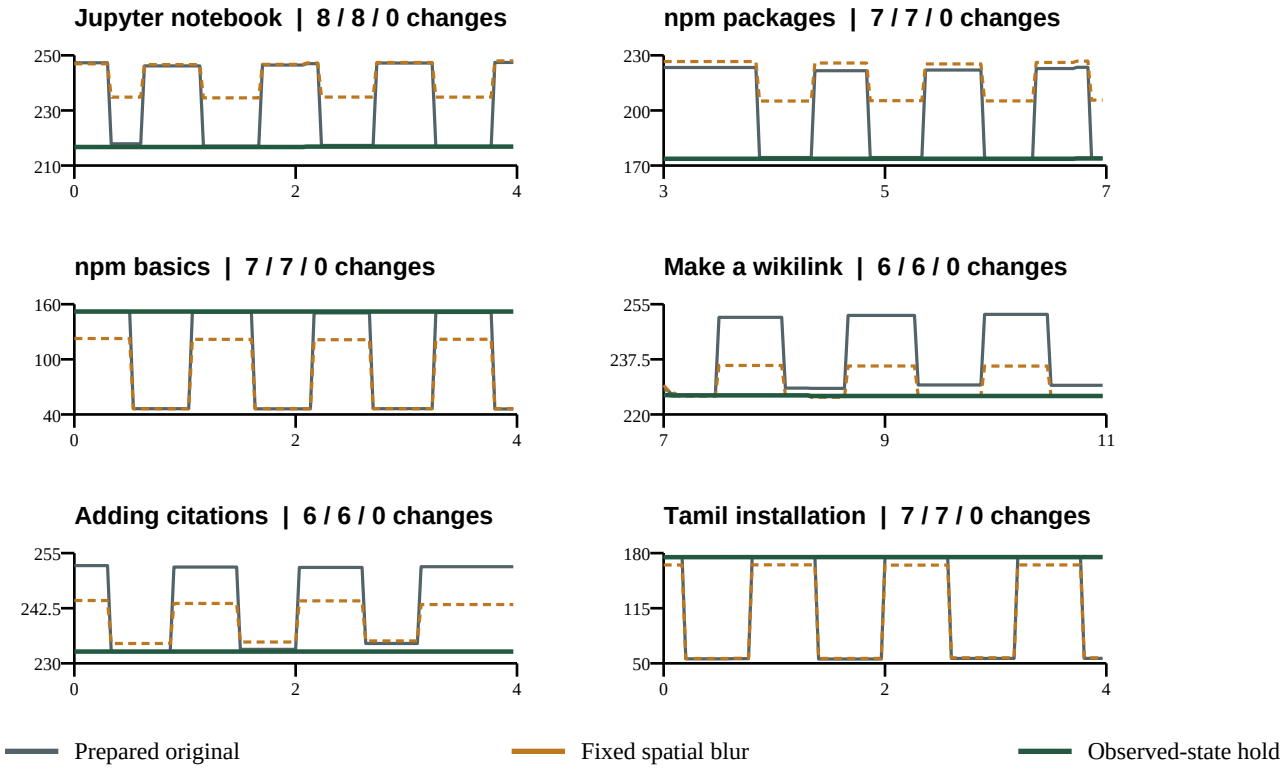
Real failure	Causal correction
Compressed browser carets failed older range/error thresholds	Proposal v2: range >=24, separation >=12, adaptive mean error max(2,min(6,0.15 x separation)); native sample bound 16 stays. Exact encoded gates stay.
A stopped or moved caret left a long inactive tail	Every state run, including first/end, is capped at 60 frames. The full citations interval is rejected; a supported four-second interval remains.
Silent VP8 preparation produced center-sited chroma	Convert chroma location to left and probe the prepared stream. The real silent lesson now reaches review/no-edit recovery.
Jupyter result linked the terminal credit	Bind its provenance link to validated example identity; verify actual downloaded credit against input bytes. UTF-8 headers fix corrupted punctuation.

The authored silent-input regression initially assumed an explicit VP8 center tag that FFmpeg did not persist. That test assumption was corrected; strict prepared-left and actual-engine acceptance remain required. Both the real runtime failure and development test failure are retained.

Six actual encoded comparisons

Six natural lesson sources / actual encoded measurements

ROI mean RGB vs. excerpt time (seconds); each panel has its own labelled RGB range



Eight-code adjacent-change threshold. Entry/return excluded. No measured learning or creator benefit.

Natural source	Excerpt start	Frames	Source / blur / hold
Jupyter notebook	301 s	0-119	8 / 8 / 0
npm packages	29 s	90-209	7 / 7 / 0
npm basics	232 s	0-119	7 / 7 / 0
Make a wikilink	60 s	210-329	6 / 6 / 0
Adding citations	24 s	0-119	6 / 6 / 0
Tamil installation	68 s	0-119	7 / 7 / 0

A declared reference. The fixed spatial blur is a 5x5 Gaussian, sigma 2, inside the same native luma box and interval, retaining source chroma. Its actual encoded output was decoded and checked. This is one untuned reference, not a complete professional editor or user trial. Continuous peak-to-peak values, adjacent RMS and reference-state contrast are published with all traces.

Jupyter ROI-mean peak-to-peak range: source 30.542, blur 13.417, hold 0.146 RGB codes. Kept reference-state contrast against its ring: source/hold 30.333, blur 12.333. This measures local contrast, not readability. Exact native held samples can still have small decoded RGB variation because interpolation depends on unchanged neighbours (largest hold range here: 0.306 codes).

A real tradeoff. The hold removes interval-interior changes but may create a worse return. Jupyter's limited selected-mean exit diagnostic rises from approximately 0.00021 to 0.23028; the UI explicitly warns that a strong return was introduced. All six boundaries remain in the receipts. Counts and exact teaching pixels do not establish good motion, legibility, safety or publishing clearance.

Selection, negatives and provenance

All selected outcomes remain. Four new tutorial sources were chosen by title, with twelve-second windows at quarter, half and three-quarter duration, floored and capped at 600 seconds. The timing plan preceded outcome inspection. Final code was then revised using development cases, so this is transparent development evidence rather than a held-out benchmark. There is no exhaustive annotation, representative sampling or detector-recall estimate.

New source	Fixed starts (seconds)	Proposal / qualified HTTP repair
Jupyter notebook	150 / 301 / 452	Only 301 proposed; that original upload qualifies.
npm packages	29 / 58 / 87	Only 29 proposed; that original upload qualifies.
npm basics	232 / 465 / 600	Only 232 proposed; that original upload qualifies.
Cangjie screencast	69 / 139 / 209	No proposals. Original 139 separately passes no-edit recovery.

The two earlier wikilink/citations sources are development repairs, not newly independent corpus additions. Tamil is the existing qualified source repeated with the current first proposal. Prepared presets, local reruns and cloud repeats do not increase the number of sources. The earlier six-source/fourteen-excerpt rapid-retimer zero-success audit, Blender teaching-content refusal and camera-lamp recurrence failures remain published. The added-corner classroom challenge remains a separately disclosed controlled positive.

Media credits and changes. The three new spoken tutorials are listed as CC BY 3.0 on Wikimedia Commons; their imported YouTube license reviews are marked not yet reviewed. Source pages, original hashes/byte counts and original URLs are published. Those three full originals are not redistributed in this package; two small prepared English examples and their credits are bundled. No endorsement is implied.

Source / credited author	License / provenance
Installing Jupyter Notebooks-Anaconda - Python for Beginners.webm / Alex The Analyst	CC BY 3.0
Npm explainer tutorial video - Expand Your Project with External Packages - Managing Package / Ganesh H	CC BY 3.0
Npm for absolute beginners (Node.JS, Visual Studio Code).webm / Kevin Powell	CC BY 3.0
How to make a wikilink / Psypherium	CC BY 3.0
Tamil installation tutorial (SimpleScreenRecorder) / Info-farmer	CC0 1.0
8.2 Adding citations / Jimmyjrg (James Gaunt), with embedded Wikimedia Foundation material	CC BY 4.0; embedded Wikimedia material CC BY-SA 3.0

Adding citations: Jimmyjrg (James Gaunt), with embedded Wikimedia Foundation material. Original CC BY 4.0 / embedded CC BY-SA 3.0; distribute any adapted lesson media under CC BY-SA 4.0 with both credits. Wikilink: Psypherium, CC BY 3.0. Tamil: Info-farmer, CC0. Cangjie: OnionBulb, screenshot license metadata recorded; original not bundled. All preparations extract, resize/letterbox and re-encode; local edits hold the reviewed observed state.

Full source pages and hashes: github.com/josepha-mayo/opencv-frame-review/blob/main/evidence/natural-v4/sources.json

Run it. Check the downloaded result.

Live AWS execution. https://le-d72c965de52749b08fd72a2513bccce1.ecs.us-east-1.on.aws
The completed ECS/Fargate rollout runs one worker/task behind managed HTTPS, with digest-pinned ECR and CloudWatch Logs. Untouched Jupyter and npm basics originals complete browser upload, preparation, review, explicit approval, export and hash-checked downloads. Their outputs match independently decoded local qualifications. All twelve browser crop hashes, completed-job reload, owner isolation and downloaded twelve-second playback pass. Listening and comprehension were not evaluated.

Quick judge route. Choose Jupyter notebook / English, start its real example, inspect both observed states, and add code/output [52,96,552,45] plus controls [50,30,560,60]. Check the plan and explicitly generate a comparison. Inspect 8 -> 0 internal changes, all twelve crops, exact teaching/audio results and the strong-return warning. npm and Tamil offer two other attributed examples. For ordinary intake, use the hash-checked original and the recorded excerpt start; source and prepared downloads are separate.

Reproduce on Linux. Build the root Dockerfile: `docker build -t frame-review .`
Run `docker run --name frame-review-check --network none frame-review python studio/run.py --verify /tmp/qualification` and copy its output with `docker cp frame-review-check:/tmp/qualification ./qualification`. Use new names. Run the loopback studio with host networking and `python studio/run.py --serve --port 8877`. Exact commands are in README. All thirteen stages pass, including 137 backend tests, studio tests (one platform skip), three natural preset exports, silent WebM transport and media/refusal suites. Windows source/UI checks are separate.

Limitations and responsible operation. Intake is capped at 64 MiB and supports bounded eight-bit SDR video with at most one mono/stereo audio track. It prepares only 1-12 seconds to 640x360/30 fps H.264 and optional AAC-LC mono/48kHz. Preparation changes picture/audio and may erase small labels. No full-lesson splice-back, semantic distraction understanding or object tracking exists. The exact edit reference is prepared media, not the uploaded original. The teacher must check readability and both boundaries.

The worker runs as UID 10001 with process/time/memory/disk caps and file/pipe protocols. Secure/HttpOnly/SameSite cookies and host/origin/request-token checks scope jobs. No application AWS API role is needed. Jobs expire after 24 hours of inactivity, and restart/deployment can clear them sooner. Known multimedia package findings remain; these restrictions are mitigations, not certification.

Competition fit. Technical depth is in OpenCV and encoded preservation; innovation is the reviewable observed-state action. Impact evidence is six natural-source repairs plus measured references; user benefit remains unmeasured. Three natural examples and refusal recovery support UX. The source/report/scripts expose negative outcomes. AWS runs the media workload. No COOL or autonomous-agent award claim is made. The existing 2:46 video shows an earlier workflow with disclosed neural narration; Joseph's final real-voice recording remains deferred.

Runtime files: 103
Manifest SHA-256: 21fb29c79616e9afebdcfc0176e4ddd44c3be83b3b58386793b9d2623f4a93d0
Image digest: sha256:196b508dcd7314e26f5d69d0fd82470821793f6de3c437d14bb7448da9d93eaf
Code, complete receipts and reproduction: https://github.com/josepha-mayo/opencv-frame-review